CS5234 Mock Quiz 1

20 pts · 50 min

20 points · 50 minutes · cheat sheet allowed　20 分 · 50 分钟 · 可带 cheat sheet

Part A · Mock Quiz (20 pts)

Problem 1 (7 pts) · Estimating a sum

You have query access to an array $a_1, \ldots, a_n$ with every $a_i \in \{1, 2, \ldots, M\}$. Let $S = \sum_i a_i$.
可查询数组 $a_1, \ldots, a_n$，每个 $a_i \in \{1, 2, \ldots, M\}$。令 $S = \sum_i a_i$。

(a) [2] Give an estimator X that uses one query and satisfies $\mathbb{E}[X] = S$.
设计一个只用 1 次查询、满足 $\mathbb{E}[X] = S$ 的估计量 X。

(b) [3] Show $\mathrm{Var}(X) \le MS^2$. Conclude that averaging $k = O(M/\varepsilon^2)$ independent copies gives $\Pr[|\bar{X} - S| \le \varepsilon S] \ge 2/3$. Hint: bound $\mathbb{E}[X^2]$ using $a_i \le M$, then use $S \ge n$.
证明 $\mathrm{Var}(X) \le MS^2$，并说明取 $k = O(M/\varepsilon^2)$ 份独立副本的平均即可使 $\Pr[|\bar{X} - S| \le \varepsilon S] \ge 2/3$。提示：用 $a_i \le M$ 上界 $\mathbb{E}[X^2]$，再用 $S \ge n$。

(c) [2] How do you boost the success probability to $1 - \delta$? What is the total number of queries?
如何把成功率提高到 $1 - \delta$？总查询次数是多少？

Problem 2 (7 pts) · Zero ones or k ones

The input $x \in \{0,1\}^n$ is promised to be either all zeros, or to contain exactly k ones ($1 \le k \le n/2$). Decide which, with success probability at least $2/3$.
输入 $x \in \{0,1\}^n$ 保证要么全为 0，要么恰有 k 个 1（$1 \le k \le n/2$）。判断是哪一种，成功率至少 $2/3$。

(a) [2] Give a randomized algorithm with $O(n/k)$ queries and analyse it.
给出 $O(n/k)$ 次查询的随机算法并分析。

(b) [3] Let μ_0 be the all-zero string and μ_1 put ones on a uniformly random k-subset. For any deterministic algorithm making q queries, let ν_0, ν_1 be the distributions of its query–answer transcript under μ_0, μ_1. Prove $\Delta_{\mathrm{TV}}(\nu_0, \nu_1) \le qk/n$.
令 μ_0 为全 0，μ_1 在均匀随机的 k 元子集上放 1。对任意 q 次查询的确定性算法，令 ν_0, ν_1 为两种情况下查询-回答记录的分布。证明 $\Delta_{\mathrm{TV}}(\nu_0, \nu_1) \le qk/n$。

(c) [2] Use Yao's principle to conclude $R(f) = \Omega(n/k)$. What does this say about testing NotOR (take $k = \varepsilon n$)?
用 Yao 原理得出 $R(f) = \Omega(n/k)$。这对 NotOR 性质测试（取 $k = \varepsilon n$）意味着什么？

Problem 3 (6 pts) · Streaming sample of size k

A stream $x_1, \ldots, x_n \in [m]$ arrives one element at a time; n is unknown in advance. Algorithm: store the first k elements; when the i-th element arrives ($i > k$), with probability k/i it replaces a uniformly random one of the k stored elements.
数据流 $x_1, \ldots, x_n \in [m]$ 逐个到达，n 事先未知。算法：存下前 k 个元素；第 i 个元素（$i > k$）到达时，以概率 k/i 替换当前 k 个样本中均匀随机的一个。

(a) [3] Prove by induction that after processing element i ($i \ge k$), every x_j with $j \le i$ is in the sample with probability exactly k/i.
用归纳法证明：处理完第 i 个元素后（$i \ge k$），每个 x_j（$j \le i$）在样本中的概率恰为 k/i。

(b) [1] How much space does the algorithm use?
算法需要多少空间？

(c) [2] Let p be the fraction of even numbers in the stream, estimated by the fraction $\hat{p}$ of even numbers in the final sample. You may use that the indicators "x_j is in the sample" are pairwise non-positively correlated. Show that $k = O(1/\varepsilon^2)$ gives $\Pr[|\hat{p} - p| \le \varepsilon] \ge 2/3$.

设流中偶数比例为 p，用最终样本中偶数的比例 $\hat{p}$ 估计。可以直接使用「x_j 在样本中」这些指示变量两两非正相关。证明 $k = O(1/\varepsilon^2)$ 时 $\Pr[|\hat{p} - p| \le \varepsilon] \ge 2/3$。

Part B • Extra practice

E1 • Maximum load (L1)

Throw n balls independently and uniformly into n bins. (a) Show that with probability $1 - o(1)$ no bin receives more than $10 \ln n$ balls. (b) Why does Chebyshev's inequality not suffice here?

n 个球独立均匀地投入 n 个箱。(a) 证明以 $1 - o(1)$ 的概率，没有箱子收到超过 $10 \ln n$ 个球。(b) 为什么这里用 Chebyshev 不够?

E2 • Counting with relative error (L2)

An n-bit string has A ones, and you are promised $A \ge t$ for a known t. (a) Give a $(1 \pm \varepsilon)$-approximation of A with probability $2/3$ using $O\left(\frac{n}{t\varepsilon^2}\right)$ queries. (b) Sketch how to proceed if t is unknown (but $A \ge 1$).

n 位串中有 A 个 1，已知 $A \ge t$（t 已知）。(a) 用 $O\left(\frac{n}{t\varepsilon^2}\right)$ 次查询以 $2/3$ 的概率给出 A 的 $(1 \pm \varepsilon)$ 近似。(b) 若 t 未知（但 $A \ge 1$），给出做法思路。

E3 • Testing whether a string is constant (L4)

Property: all bits of $x \in \{0,1\}^n$ are equal. (a) What is the distance of x from the property? (b) Give a one-sided tester with $O(1/\varepsilon)$ queries and prove it works.

性质：$x \in \{0,1\}^n$ 的所有位都相同。(a) x 到该性质的距离是多少？(b) 给出一个 $O(1/\varepsilon)$ 次查询的单边测试器并证明其正确。

E4 • Triangle detection needs $\Omega(n^2)$ (L3)

With pair queries on an n-vertex graph, prove that deciding whether the graph contains a triangle has randomized query complexity $\Omega(n^2)$. Hint: reduce from OR; add one special vertex.

在 n 个顶点的图上用 pair query，证明判断图中是否有三角形的随机查询复杂度为 $\Omega(n^2)$。提示：从 OR 归约，加一个特殊顶点。